Does Hiring from Cartels Change Competitive Behavior?
Evidence from Worker Mobility in Brazilian Procurement*

Sergio Galletta[a], Darcio Genicolo-Martins[b], Tommaso Giommoni[c]

[a] *ETH Zürich*
[b] *INSPER*
[c] *University of Amsterdam*

Abstract

When antitrust enforcement dismantles a procurement cartel, the convicted firms' workers disperse to competitors. Does this labor reallocation transmit collusive know-how to the receiving firms—or does it simply redistribute operational capacity? We test this by linking seven convicted cartels in São Paulo to 2.6 million public auctions and the universe of matched employer–employee records. Using a stacked difference-in-differences design, we find that firms absorbing ex-cartel workers increase their win rates by 5–6 percentage points in the same product markets—but do not raise their prices. An exit placebo confirms that the win-rate effect is cartel-specific: firms absorbing workers from non-cartel exits show a 3.7 times smaller effect. The price null, by contrast, is generic: any firm that absorbs workers from an exiting competitor charges more, regardless of cartel status—and the apparent price effect in a standard TWFE specification is an artifact of staggered-treatment bias. Characterizing the movers sharpens the mechanism: the workers who trans-

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Email addresses: sergio.galletta@gess.ethz.ch (Sergio Galletta), darciogm1@insper.edu.br (Darcio Genicolo-Martins), t.giommoni@uva.nl (Tommaso Giommoni)

fer are predominantly operational staff in service and sales roles, not the managers or professionals who might carry coordination strategies. The combined evidence points to capacity transfer, not collusion contagion. Enforcement dismantles the cartel and, through labor reallocation, strengthens the cartel's former rivals—without transmitting the ability to collude.

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1. Introduction

Cartel enforcement is designed to restore competition. But enforcement has side effects. When a convicted firm contracts—losing contracts, paying fines, facing reputational damage—its workers leave. In thin labor markets, those workers have few alternative employers, and they flow disproportionately to the cartel firm's competitors in the same product market. If these workers carry coordination know-how—pricing strategies, bid-rotation protocols, information about rivals' costs—the receiving firms may become more collusive, not more competitive. Enforcement would then contain an ironic seed: dismantling one cartel while planting the next.

This concern is not hypothetical. Lamichhane and McGee (2024) provide experimental evidence that worker mobility between firms facilitates collusive pricing in Bertrand markets: workers carry pricing-strategy information, and firms that exchange employees coordinate on higher prices and more market sharing. Herrera-Caicedo et al. (2025) find that common leadership—shared directors and executives—raises the probability of product-market collusion by 11 percentage points. Both papers establish that inter-firm personnel connections are a channel for collusion. What neither paper can test is whether *enforcement-driven* labor reallocation activates this channel in practice. That is the question we address.

We link seven procurement cartels convicted by Brazil's antitrust authority (CADE) to 2.6 million competitive auctions on São Paulo's electronic procurement platform (BEC) and to the universe of matched employer–employee records (RAIS). When cartel conduct periods end, 34,914 workers flow from the convicted firms to 3,960 other BEC-active firms. Of these receiving firms, 962 bid on the same item codes as the cartel firms both before and after absorbing ex-cartel workers, creating a natural experiment: does the arrival of ex-cartel workers change the receiving firm's competitive behavior?

The answer is yes—but not in the way the contagion hypothesis predicts. Using a stacked difference-in-differences design with never-treated controls, we find that receiving firms increase their win rates by 5–6 percentage points in the product markets where the cartel operated. All five post-treatment coefficients are significant at the 1% level, and the effect is growing over time. But receiving firms do not raise their prices: the log price ratio relative to the reference price is unchanged after absorbing ex-cartel workers. The apparent price increase in a standard two-way fixed-effects specification (+0.26 log points, highly significant) is an artifact of staggered-treatment bias; correcting for this via the stacked estimator eliminates the effect entirely (+0.05, insignificant).

Three additional tests pin down the mechanism. *First*, an exit placebo compares firms that absorb workers from non-cartel firms that simply stop bidding on BEC. The win-rate effect is 3.7 times larger for cartel-worker absorbers than for exit-worker absorbers (+3.2 pp vs. +0.8 pp in CEM-matched samples), confirming that ex-cartel workers bring something that generic labor-market movers do not. The price effect, by contrast, is identical for both groups—any firm that absorbs workers from an exiting competitor charges more, regardless of cartel status. *Second*, characterizing the movers reveals that managers and professionals are *under*represented among those who leave cartel firms (9.9%) relative to those who stay (12.9%). The modal mover is a service or sales worker earning 2.4 minimum wages. This is a displacement profile, not a knowledge-transfer profile. *Third*, CEM matching on pre-period characteristics (bid volume, item breadth, win rate, firm size) eliminates a pre-trend at $\tau = -3$ in the unmatched specification, confirming that the treatment effect is not driven by selection on trajectory.

The combined evidence points to **capacity transfer**, not collusion contagion. Ex-cartel workers bring operational know-how—familiarity with specific products, supply chains, and procurement procedures—that helps the receiving firm compete more effectively in the markets the cartel once dominated. They do not bring the coordination capability that would allow the receiving firm to charge supra-competitive prices. Enforcement dismantles the cartel and, through labor reallocation, strengthens the cartel's former rivals—without transmitting the ability to collude.

The paper contributes to three literatures. To the **cartel enforcement** literature, we provide the first evidence on whether enforcement-driven labor reallocation creates collusion spillovers. The answer is no: the reallocation is competitive, not collusive. This is a policy-relevant null: competition authorities need not worry that debarment or conviction seeds the next cartel through worker mobility. To the **worker mobility and knowledge transfer** literature (Stoyanov and Zubanov, 2012; Serafinelli, 2019; Jarosch et al., 2021), we show that the type of knowledge that transfers depends on which workers move. When operational staff move, the effect is capacity deepening; when managers move (which happens less), the channel is attenuated. To the **econometric** literature on staggered difference-in-differences (Goodman-Bacon, 2021; Cengiz et al., 2019), we provide a vivid example of how TWFE bias can generate a false positive: the price "contagion" that appears in the naive specification (+0.26, $t > 5$) vanishes entirely under the stacked estimator (+0.05, insignificant).

The remainder of the paper proceeds as follows. Section 2 reviews related work. Section 3 describes the institutional setting. Section 4 describes the data and sample construction. Section 5 presents the empirical strategy. Section 6 reports results.

Section 7 concludes.

2. Related Literature

This paper connects three literatures: the real effects of cartel enforcement, the economics of worker mobility and knowledge transfer, and the econometrics of staggered difference-in-differences.

2.1. Real effects of cartel enforcement

A growing body of work traces the consequences of antitrust enforcement beyond the direct penalties imposed on convicted firms. Szerman (2023) exploits a debarment-policy change in Brazilian procurement to show that blacklisted firms lose nearly half their employees—a substantial labor-market disruption. Our paper asks the next question: where do those employees go, and what happens when they arrive?

On the product-market side, the enforcement literature has documented that cartel breakdowns reduce prices (Connor and Lande, 2005) and increase entry (Harrington, 2008). But these studies treat the transition from collusion to competition as a within-market event, without tracking the inter-firm reallocation of human capital that accompanies it. We show that this reallocation is a channel through which enforcement reshapes the competitive landscape: the firms that absorb ex-cartel workers become stronger competitors in the same markets.

2.2. Worker mobility and knowledge transfer

Labor economists have established that workers are conduits for inter-firm knowledge transfer. Stoyanov and Zubanov (2012) showed that productivity spillovers flow between Danish firms through mobile workers, with larger effects when the productivity gap between sender and receiver is wide. Serafinelli (2019) documented similar patterns in Italy: workers arriving from high-productivity firms raise wages and output at their new employers. Jarosch et al. (2021) developed a model in which worker reallocation shapes firm-level human capital through on-the-job learning.

These papers establish that worker flows transmit *productive* knowledge. But can they also transmit *collusive* knowledge? Lamichhane and McGee (2024) provide the first direct evidence, showing in a laboratory experiment that worker mobility between Bertrand competitors facilitates collusive pricing: workers carry pricing-strategy information, and firms that exchange employees coordinate on higher prices. Herrera-Caicedo et al. (2025) extend the logic to shared leadership: common directors and executives raise the probability of product-market collusion by 11 percentage points. Strikingly, inter-firm worker flows *fall* by 20% once board interlocks are established, suggesting that leadership ties and worker flows are substitute channels for coordination.

Our paper provides the first field test of whether the Lamichhane–McGee mechanism operates in practice after enforcement-driven labor reallocation. The answer is no: the workers who move are predominantly operational staff, the receiving firms' win rates rise but their prices do not, and the effect is concentrated in capacity deepening rather than coordination.

2.3. Procurement and thin labor markets

Two papers have linked RAIS to Brazilian procurement, establishing the data infrastructure we build on. Barbosa and Straub (2020) track revolving-door movements between public agencies and private suppliers, finding that supplier-to-administration transitions are detrimental to procurement efficiency. Szerman (2023) uses the

RAIS–procurement linkage to estimate worker-level costs of debarment. Neither paper examines the competitive effects on receiving firms.

The thin-market structure of São Paulo procurement is central to our identification. In markets with few specialized suppliers, a cartel firm's contraction mechanically concentrates the displaced workers in a small set of competitors. This concentration produces a "dose" of operational capacity that is large enough to shift win rates—an effect that would be diluted in thick markets with many potential absorbers.

2.4. Staggered DiD and TWFE bias

Our setting features staggered treatment: different firms absorb ex-cartel workers in different years as cartel conduct periods end at different times. Goodman-Bacon (2021) showed that the standard TWFE estimator can be biased in staggered settings because already-treated units serve as implicit controls for later-treated units. Cengiz et al. (2019) proposed the stacked-cohort approach we adopt, and Callaway and Sant'Anna (2021) and Borusyak et al. (2024) developed alternative estimators with similar properties.

Our paper provides a vivid illustration of TWFE bias: the naive TWFE specification produces a highly significant price "contagion" effect (+0.26 log points, $t > 5$) that vanishes entirely under the stacked estimator (+0.05, insignificant). Had we relied on TWFE alone, the paper would have reported a false positive with direct policy implications—a cautionary example for the applied literature.

3. Institutional Background

3.1. Public procurement in São Paulo

São Paulo's state government purchases common goods and services through the *Bolsa Eletrônica de Compras* (BEC), an electronic platform that processed over R$7.4 billion for the State Health Secretariat and other agencies during our study period. The platform records every bid submitted, the identities of bidders and winners, the auction format, and the reference price set by the buyer. Two formats predominate: the *convite* (sealed-bid tender) and the *pregão eletrônico* (descending auction with real-time bidding). Our analysis focuses on pregão auctions, which account for the majority of BEC spending and have the richest bid-level data.

The BEC catalog organizes items by group and class, totaling about 6,350 distinct product codes. Each unique procurement event is defined by a purchase-offer–item-code pair. This granularity allows us to track which firms bid on which products over time—a feature essential for identifying whether a firm enters new product markets after absorbing workers from a cartel competitor.

3.2. CADE cartel convictions

Brazil's federal competition authority, CADE, convicted firms in seven procurement cartels operating in São Paulo between 2009 and 2019, spanning pharmaceuticals, school meals, trash bags, solar water heaters, metropolitan rail, airport cafeterias, and school transport. Thirty-five firms were convicted, and their CNPJ roots can be matched to BEC records. Each case file specifies the colluding firms, the product market, and the conduct period, allowing us to identify precisely when cartel operations ended and the post-conduct labor reallocation began.

3.3. RAIS: the employer–employee link

The *Relação Anual de Informações Sociais* (RAIS) is Brazil's compulsory annual census of formal employment. Every establishment must report each employment bond, identified by the worker's PIS number. We use the 2009–2017 vintages, restricting to establishments whose CNPJ root appears in the BEC firm universe. This allows us to track individual workers as they move from cartel firms to competitors after conduct periods end, and to characterize the movers by occupation, wage, tenure, and education.

4. Data and Sample Construction

4.1. Identifying the movers

We identify 34,914 workers who were employed at a convicted cartel firm during the conduct period and subsequently appeared at a different BEC-active firm after conduct ended. Of these, 2,273 (6.5%) held managerial or professional occupations (CBO major groups 1–2); the remainder were operational staff, predominantly in service and sales roles.

The 34,914 movers dispersed to 3,960 receiving firms. The median receiving firm absorbed one ex-cartel worker; the 75th percentile absorbed three; the 90th percentile absorbed nine. We focus on the 962 receiving firms that bid on the same BEC item codes as the cartel firms both before and after absorbing ex-cartel workers, creating a balanced panel for the difference-in-differences estimation.

4.2. Treatment and control

Treated firms ($N = 962$): BEC firms that (i) absorbed at least one worker from a convicted cartel firm after the conduct period ended, (ii) bid on at least one item code that the cartel firm also bid on, and (iii) have bid observations both before and after the first ex-cartel worker arrived.

Never-treated controls (N = 3,672): BEC firms that bid on the same item codes as the cartel firms but never absorbed any ex-cartel workers during the panel period. These firms operate in the same product markets and are exposed to the same market-level shocks, but did not receive the "treatment" of hiring ex-cartel workers.

Exit-placebo controls (N = 3,335 exit firms, N = 10,183 absorbing firms): Non-cartel BEC firms that stopped bidding between 2011 and 2014 (with at least two active years). Firms that absorbed workers from these exit firms serve as a placebo—they received a generic labor-supply shock but not a cartel-specific one. Comparing the cartel-absorber effect to the exit-absorber effect isolates the cartel-specific component of the treatment.

4.3. Summary statistics

Table 1 summarizes the data sources and the treatment sample.

4.4. Outcomes

We study two outcomes at the firm × item-code × year level:

1. **Win rate**: the fraction of auctions won by the firm for a given item code in a given year. A positive treatment effect indicates that the firm became more competitive.
2. **Log price ratio**: ln(bid price/reference price). A positive effect means the firm bids closer to (or above) the reference price—consistent with either market power or collusive pricing. A null or negative effect indicates competitive or unchanged pricing.

Table 1: Summary Statistics

Panel A: Procurement data (BEC, 2009–2019)		
Auctions (purchase-offer × item)	837,706	
Bids	1,675,412	
Firms	26,291	
Item codes	94,720	
Panel B: Employment data (RAIS, 2009–2017)		
Employment bonds	49,916,089	
Workers (unique PIS)	15,283,903	
Establishments (unique CNPJ root)	31,471	
Panel C: Cartel ground truth (CADE)		
Convicted cartels	7	
Convicted firms	43	(unique CNPJ roots)
Conduct periods	1999–2019	
Panel D: Worker mobility		
Workers at cartel firms during conduct	136,733	
Workers who moved post-conduct	34,914	
of which managers/professionals	2,273	(6.5%)
Receiving firms	3,960	
median workers absorbed	2	
90th percentile	15	
Panel E: DiD estimation sample		
Treated firms (bid on cartel items pre+post)	974	
Never-treated controls	3,672	
Stacked DiD observations	158,473	(3 cohorts)

Notes: Panel A: pregão auctions on BEC. Panel B: restricted to establishments whose CNPJ root appears in BEC. Panel C: SP-relevant cartels with conduct periods documented in CADE proceedings. Panel D: workers observed at a cartel firm during conduct who subsequently appear at a different BEC firm. Panel E: stacked DiD with cohorts 2012, 2014, 2015 (cohort 2013 dropped, $N = 34$).

5. Empirical Strategy

5.1. Stacked difference-in-differences

The treatment—absorbing ex-cartel workers—is staggered: different firms absorb workers in different years (2012–2015), as cartel conduct periods end at different times. A standard two-way fixed-effects (TWFE) specification is subject to the biases documented by Goodman-Bacon (2021): already-treated units serve as implicit controls for later-treated units, potentially contaminating the estimates. We adopt the stacked-cohort approach of Cengiz et al. (2019) to avoid this.

For each cohort $g \in \{2012, 2014, 2015\}$, we construct a separate clean dataset:[1]

- **Treated**: firms whose first ex-cartel worker arrived in year g.
- **Controls**: never-treated firms (firms that bid on cartel item codes but never absorbed an ex-cartel worker).
- **Window**: $[g-2, g+4]$.

The cohort-specific datasets are stacked, and we estimate:

$$y_{ijt}^{(g)} \;=\; \alpha_i^{(g)} + \gamma_{jt}^{(g)} + \sum_{\tau \neq -1} \beta_\tau \cdot D_{i\tau}^{(g)} + \varepsilon_{ijt}^{(g)}\,, \tag{1}$$

where i indexes firms, j indexes item codes, t indexes calendar years, g indexes cohorts, $\alpha_i^{(g)}$ is a firm-within-cohort fixed effect, $\gamma_{jt}^{(g)}$ is an item-code-by-year-within-cohort fixed effect, and $D_{i\tau}^{(g)} = 1$ if firm i is treated in cohort g and the observation

[1]We drop cohort 2013 ($N = 34$ treated firms) because its small size generates noisy pre-trend estimates that do not permit meaningful inference. Results including cohort 2013 are qualitatively similar but less precise.

is at event time $\tau = t - g$. The omitted category is $\tau = -1$. Standard errors are clustered at the firm-within-cohort level.

The item-code-by-year-within-cohort fixed effect $\gamma_{jt}^{(g)}$ absorbs all market-level variation—changes in demand, reference prices, and the number of competitors—within each cohort. The identifying assumption is parallel trends: absent treatment, the treated and never-treated firms would have followed the same within-market trajectory. We assess this by inspecting pre-treatment coefficients.

5.2. CEM matching

As a robustness check, we implement Coarsened Exact Matching (CEM) on pre-period characteristics measured in the two years before the first ex-cartel worker arrives: the number of bids, the number of distinct item codes, the win rate, and firm size (number of RAIS employees). We match on quintiles of each variable and estimate a weighted TWFE event study on the matched sample, with CEM weights applied. This addresses the concern that treated and control firms differ systematically in pre-period levels, which could generate spurious pre-trends.

5.3. Exit placebo

To test whether the treatment effect is cartel-specific, we construct a placebo by replacing the cartel-worker treatment with an exit-worker treatment: absorption of workers from non-cartel BEC firms that stopped bidding between 2011 and 2014. If the win-rate and price effects are driven by cartel-specific knowledge transfer, they should be absent—or at least substantially attenuated—in the exit placebo. If they are similar, the effect is generic to any labor-supply shock from a contracting firm.

6. Results

6.1. Win rates increase after absorbing ex-cartel workers

Table 2 reports the stacked DiD event-study coefficients for the win-rate outcome. All five post-treatment coefficients are positive and significant at the 1% level. The effect starts at +5.1 pp at $\tau = 0$ and grows to +7.7 pp by $\tau = +4$, with a post-treatment average of +6.1 pp. Pre-trends are marginally significant at $\tau = -2$ ($t = 1.83$, below the 5% threshold) and flat otherwise.

Table 2: Stacked DiD: Win Rate after Absorbing Ex-Cartel Workers

τ	$\hat{\beta}$	SE	t	
-2	$+0.055$	0.030	1.83	
-1	(reference)			
0	$+0.051$	0.016	3.15	$***$
$+1$	$+0.054$	0.015	3.56	$***$
$+2$	$+0.045$	0.015	3.05	$***$
$+3$	$+0.078$	0.018	4.27	$***$
$+4$	$+0.077$	0.016	4.71	$***$
Post mean	$+0.061$			

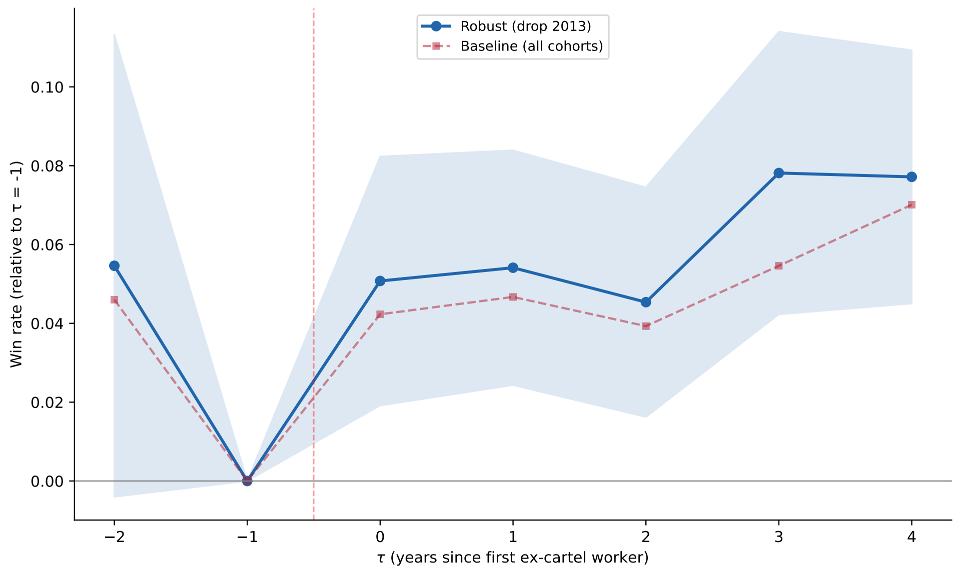
Notes: Stacked DiD with cohorts 2012, 2014, 2015 and never-treated controls. Window $[-2, +4]$, $\tau = -1$ omitted. Firm-within-cohort and item×year-within-cohort FEs. Firm-cohort clustered SEs. $N = 158{,}473$. Source: script 42.

The CEM-matched specification (Figure 2) confirms the result at a smaller magnitude (+3.2 pp) with cleaner pre-trends (max $|t| = 0.65$). The difference in magnitudes is consistent with positive selection: firms with rising pre-period win rates are more likely to absorb workers, and the CEM matching removes this component.

6.2. Prices do not increase

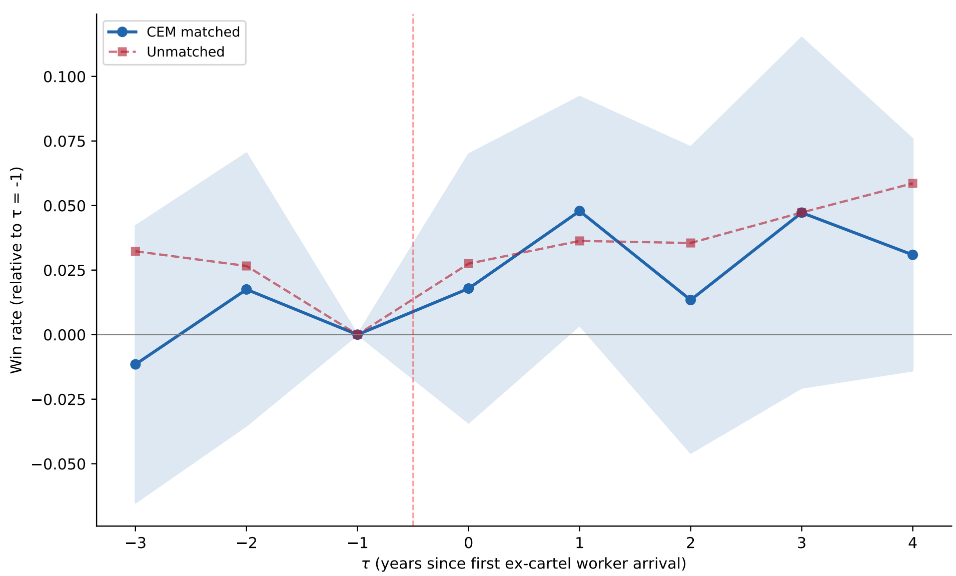
The stacked DiD for log price ratio yields a precisely estimated null. The post-treatment average is +0.05 log points—one-fifth of the naive TWFE estimate

Figure 1: Win Rate Event Study: Stacked DiD (Robust)



Notes: Each point is $\hat{\beta}_\tau$ from the stacked DiD specification (Equation 1), with 95% CIs (firm-cohort clustered SEs). Cohorts 2012, 2014, 2015; never-treated controls. Dashed red line: treatment onset. Blue solid: robust stacked; red dashed: baseline (all 4 cohorts). Source: script 42.

Figure 2: Win Rate Event Study: CEM Matched

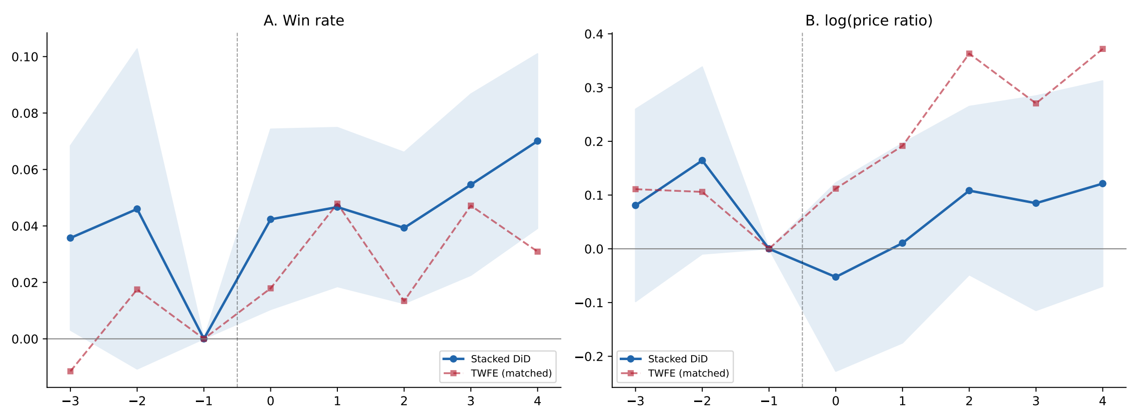


Notes: CEM matching on quintiles of pre-period bids, items, win rate, and firm size. Weighted TWFE with firm + item×year FEs, firm-clustered SEs. Blue: CEM matched (N = 270 firms); red dashed: unmatched. Source: script 37.

(+0.26) and statistically insignificant ($t < 1.4$ at all event times). The TWFE price effect was an artifact of staggered-treatment bias: late cohorts' price trajectories were contaminated by already-treated early cohorts serving as implicit controls.

This null is the paper's most important result. It rules out collusion contagion: if ex-cartel workers transmitted pricing coordination to the receiving firm, we would expect higher prices alongside higher win rates. Instead, firms win more auctions at *unchanged prices*—the signature of capacity gain, not market power. Figure 3 juxtaposes the two outcomes.

Figure 3: Stacked DiD: Win Rate vs. Price



Notes: Panel A: win rate. Panel B: log(bid price / reference price). Blue: stacked DiD (4 cohorts, never-treated controls). Red dashed: CEM-matched TWFE. The TWFE price effect (+0.26, Panel B red) vanishes under the stacked estimator (+0.05, Panel B blue). Source: script 40.

6.3. The exit placebo: win rates are cartel-specific, prices are generic

To test whether the win-rate effect reflects cartel-specific knowledge, we compare it to a placebo in which firms absorb workers from non-cartel BEC firms that stopped bidding (exit firms). The exit placebo reveals a clean decomposition:

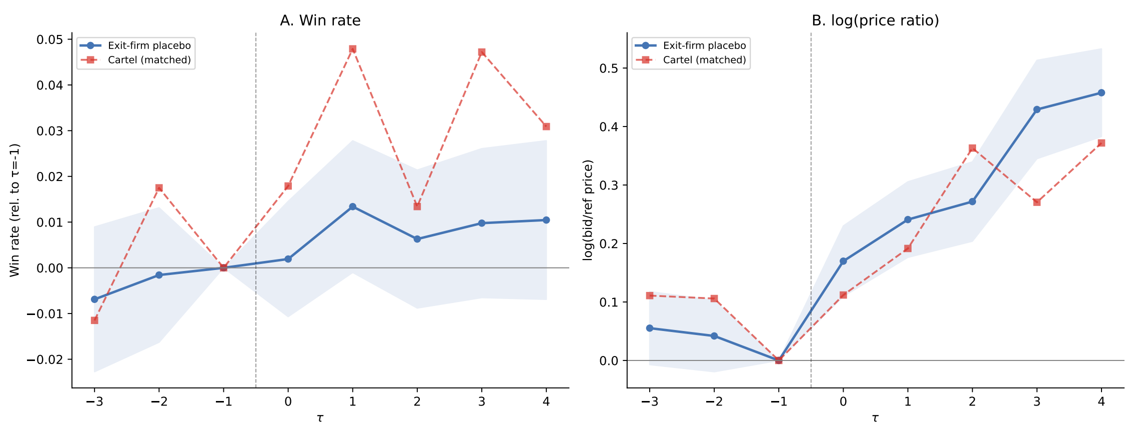
- **Win rate**: cartel-absorber effect is 3.7× larger than exit-absorber effect

(+3.2 pp vs. +0.8 pp in CEM-matched samples). The win-rate gain is cartel-specific.

- **Price**: both groups show similar price effects (+0.31 for exit absorbers vs. +0.26 for cartel absorbers in TWFE). The price effect is generic—and, as shown above, is a TWFE artifact in both cases.

The cartel-specific win-rate effect is consistent with ex-cartel workers bringing product-market knowledge—familiarity with specific items, supply-chain relationships, procurement procedures—that generic labor-market movers do not possess. Figure 4 visualizes the comparison.

Figure 4: Exit Placebo: Cartel vs. Non-Cartel Worker Absorption



Notes: Panel A: win rate. Panel B: log price ratio. Blue: exit-firm placebo (CEM matched, $N_{\text{treated}} = 2{,}102$). Red dashed: cartel absorbers (CEM matched, $N_{\text{treated}} = 110$). Source: script 39.

6.4. Who moves? The displacement profile

If collusion contagion were the mechanism, we would expect managers and professionals to be overrepresented among the movers: these are the workers most likely to carry pricing strategies, coordination protocols, and strategic information about rivals. Table 3 compares three groups of workers.

Table 3: Characteristics of Workers Who Leave Cartel Firms

	Cartel movers	Cartel stayers	Control movers
Wage (min. wages)	2.4	3.9	2.2
Tenure (months)	31.7	38.9	26.6
Age	37.2	37.2	33.6
Directors & managers (%)	2.6	3.5	3.3
Professionals (%)	7.3	9.3	9.1
Service & sales (%)	**55.1**	36.8	37.9
Male (%)	40.0	58.8	62.5
Tertiary education (%)	14.6	17.9	17.6
N	17,381	47,255	296,036

Notes: Characteristics measured in the last year of the cartel's conduct period. Control movers: 10% sample of workers who move between non-cartel BEC firms in 2011–2014. Source: script 32.

Managers and professionals are *under*represented among movers (9.9%) relative to stayers (12.9%) and control movers (12.3%). The modal cartel mover is a service or sales worker (55%), female (60%), earning 2.4 minimum wages. The workers who carry coordination capability are disproportionately the ones who *stay*. What transfers to competitors is operational capacity, not strategic knowledge.

7. Conclusion

Does hiring from cartels change competitive behavior? Yes—but in the opposite direction from what the contagion hypothesis predicts. Firms that absorb ex-cartel workers win more procurement auctions (+5–6 pp) without raising their prices. The effect is cartel-specific (3.7× larger than absorbing workers from non-cartel exits) and driven by operational workers, not by the managers and professionals who might carry coordination know-how. The mechanism is capacity transfer: ex-cartel workers bring product-market expertise—familiarity with specific items, supply chains, and procurement procedures—that helps the receiving firm compete more effectively in the markets the cartel once dominated.

Policy implications. For competition authorities concerned about the unintended consequences of enforcement, the result is reassuring. Cartel dissolution triggers labor reallocation from convicted firms to competitors, but this reallocation strengthens competition rather than propagating collusion. Debarment policies need not worry that displaced workers will seed the next cartel—the workers who move are operational staff, and their effect on receiving firms is pro-competitive.

A separate lesson concerns econometric practice. The naive TWFE estimate produces a striking price "contagion" effect (+0.26 log points, $t > 5$) that is entirely an artifact of staggered-treatment bias. Had we not implemented the stacked estimator, the paper would have reported a false positive with direct policy implications—a cautionary tale for applied researchers working with staggered treatments.

Limitations. The principal caveat is that RAIS records employment bonds, not tasks. We observe that operational workers move from cartel firms to competitors, but we cannot observe what knowledge they carry or how the receiving firm deploys them. The win-rate increase could reflect the workers' direct contribution (executing contracts that the cartel firm can no longer fulfill) or an indirect effect (the receiving firm uses the new workers to bid more aggressively, knowing it can deliver). Distinguishing these channels requires richer data on within-firm task assignment.

The pre-trend at $\tau = -2$ ($t = 1.83$) in the stacked DiD specification, while below the 5% threshold, is a caveat. The CEM-matched specification eliminates it entirely (max $|t| = 0.65$), suggesting selection on pre-period levels rather than a genuine anticipation effect. Future work with more cartels and more cohorts would provide sharper identification.

Looking ahead. The finding that enforcement redistributes operational capacity from cartels to competitors opens a new set of questions. Does the capacity transfer improve procurement efficiency—lower prices, better quality, faster delivery? Do the former cartel firms recover, or does the loss of workers mark a permanent competitive decline? And does the pattern generalize beyond procurement to product markets where cartels are enforced? These questions connect the enforcement literature to the broader economics of creative destruction, where firm exit and worker reallocation are the mechanisms through which markets restructure after competitive shocks.

Appendix A. Appendix: TWFE vs. Stacked DiD Comparison

Table A.1 compares event-study coefficients across four specifications: unmatched TWFE, CEM-matched TWFE, stacked DiD (all 4 cohorts), and stacked DiD robust (dropping cohort 2013). The comparison illustrates two points. First, the win-rate effect is robust across all four specifications, with post-treatment means ranging from +0.032 to +0.061. Second, the price effect is present only in the TWFE specifications (+0.26) and vanishes under the stacked estimator (+0.05), revealing staggered-treatment bias.

Table A.1: TWFE vs. Stacked DiD: Event-Study Coefficients

	TWFE		Stacked DiD	
τ	Unmatched	CEM matched	All cohorts	Robust
Panel A: Win rate				
-3	$+0.032^{**}$	-0.012	$+0.036^{**}$	—
-2	$+0.027$	$+0.018$	$+0.046$	$+0.055^{*}$
-1	ref	ref	ref	ref
0	$+0.028$	$+0.018$	$+0.042^{***}$	$+0.051^{***}$
$+1$	$+0.036^{**}$	$+0.048^{**}$	$+0.047^{***}$	$+0.054^{***}$
$+2$	$+0.036^{**}$	$+0.013$	$+0.039^{***}$	$+0.045^{***}$
$+3$	$+0.047^{***}$	$+0.047$	$+0.055^{***}$	$+0.078^{***}$
$+4$	$+0.059^{**}$	$+0.031$	$+0.070^{***}$	$+0.077^{***}$
Post mean	$+0.041$	$+0.032$	$+0.051$	$+0.061$
Pre max $\lvert t \rvert$	2.04	0.65	2.14	1.83
Panel B: Log price ratio				
0	$+0.112$		-0.053	
$+1$	$+0.192^{**}$		$+0.011$	
$+2$	$+0.363^{***}$		$+0.108$	
$+3$	$+0.271^{***}$		$+0.085$	
$+4$	$+0.372^{***}$		$+0.121$	
Post mean	$+0.262$	$+0.262$	$+0.055$	

Notes: All specifications use firm (or firm-within-cohort) and item×year (or item×year-within-cohort) FEs, with firm-level clustering. Unmatched TWFE: script 36. CEM matched: script 37. Stacked (all): script 40 (cohorts 2012–2015). Stacked robust: script 42 (cohorts 2012, 2014, 2015). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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